

The Fluency Gap: Why 1,200 Days of Streaks Isn't Fluency

A Product Case Study on Duolingo's Retention vs. Production Gap By Anvitha Abbagani

I hit a 1,200-day streak across Spanish, French, and Greek— and still couldn't hold a real conversation. This case study investigates why, and proposes two low-effort, high-confidence interventions to shift incentives from passive recognition toward active production.

Full competitor scoring rationale, persona detail, RICE tables, wireframes, and metrics are in the Appendix.

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1. Problem

After 1,200 days and three languages, I couldn't produce basic vocabulary without a screen in front of me. I surveyed and interviewed other long-term users to check whether this was personal or structural – it was structural. Users described completing "easy or random lessons" just to protect a streak, and rated the app's real-world usefulness as "sometimes" or "rarely" helpful.

The precise mechanism, stated carefully: this isn't primarily about visual cues; most Duolingo exercise types (translation, listening, typing) don't use images at all. The actual gap is between **recognition and production**: exercises that ask a user to select, match, or translate a given prompt reward getting the *right answer*, not the *unprompted retrieval* of it. A user can translate "Yo quiero una manzana" correctly by recognizing familiar words in context, without being able to produce the sentence cold in a live conversation. XP is awarded identically either way, so the incentive never distinguishes recognition from production, regardless of whether a given exercise happens to include an image.

"It doesn't come to my brain in real life." – Nik, Japanese learner, describing the gap between getting exercises right and speaking

JTBD: When I'm in a real conversation, I want to recall and produce language without a prompt in front of me, so I can demonstrate actual proficiency, and not just performance inside the app.

2. Competitive Context

I compared Duolingo against Babbel, Busuu, and Speak on the dimensions that most directly predict whether a user can *produce* language unprompted, using explicit criteria rather than a general impression (full rubric and rationale per criterion: **Appendix A**):

Factor	What "Strong" Means	Duolingo	Babbel	Busuu	Speak
Speaking Proficiency	Requires unprompted verbal product, evaluated for accuracy	Weak ▾	Weak ▾	Weak ▾	Strong ▾
Grammar Explanations	Explicit rule teaching before/alongside practice	Mode... ▾	Strong ▾	Weak ▾	Strong ▾
Contextual Vocabulary	Vocabulary sequenced by real-world scenarios, not solely on frequency	Mode... ▾	Strong ▾	Weak ▾	Mode... ▾

Speaking Proficiency is the one dimension where Duolingo, Babbel, and Busuu all fall behind. Only Speak, built specifically around conversational production, breaks the pattern. Duolingo already leads the category on the dimensions that build daily habits, like Tone and UI, which is what makes the recognition-over-production gap worth fixing rather than abandoning the product over. The opportunity is to pair that habit-formation strength with the one mechanic no habit-first competitor has combined it with: unprompted speaking practice.

3. Proposed Solutions

Solution 1: Speaking Retention Practice. The core novelty isn't the microphone, Duolingo already has speaking exercises. It's the **removal of the on-screen target sentence**. Current speaking exercises show the expected sentence and ask the user to read or reconstruct it from visible fragments, which is still recognition. My proposal gives only a scenario prompt ("Ask for an apple") with no visible target sentence, so the user has to generate sentence structure from memory, not just pronounce a given one – with a tiered hint system (word cue → partial cue → full reveal) so difficulty degrades gracefully instead of hard-failing a user who blanks. That's the specific gap versus what exists today.

Solution 2: Speaking XP Visibility (*revised from "remove accuracy/speed" based on the engineering and product gaps identified below*). Rather than removing existing scoring, surface a **speaking-specific XP breakdown** alongside current metrics on the completion screen, so speaking effort is visible without deprioritizing accuracy or speed for users who aren't doing speaking exercises that session. Not a "spoken XP" auto-classifier – attributing XP correctly when a user starts speaking, stops, and finishes by tapping (or dictates and then edits) is a real classification problem with ambiguous edge cases, and mislabeling it would undermine the exact trust this feature is meant to build. Instead, speaking XP is only counted for exercises explicitly flagged as speaking-type at the content level, sidestepping the ambiguity entirely.

4. Metrics & What's Next

North Star: Speaking Success Rate — % of speaking tasks completed correctly without hints, since this is the most direct proxy for unprompted production rather than engagement. Full metric set and 30/60/90-day plan: **Appendix E.**

Next Steps: Phase 1: free-write prompts for intermediates with no current writing practice; Phase 2: exam-rubric-specific grading building on (not replacing) Explain My Answer; Phase 3: scenario-based vocabulary sequencing informed by the competitive gap in Section 2.

Appendix

A: Competitor Analysis (Full Rubric)

Each factor below is scored against an explicit definition of "Strong," so ratings are comparable across apps rather than based on general impression.

- **Speaking proficiency:** Does the app require and evaluate unprompted verbal production, not just accept audio?
- **Tone:** Does the register sustain engagement without undermining rigor?
- **Contextual vocabulary:** Is vocabulary sequenced for real-world use, regardless of speaking features?
- **UI:** Does the interface stay low-friction, independent of animation density?
- **Grammar explanations:** Does the app teach rules explicitly, or only correct errors after the fact?
- **Repetition:** Does repetition adapt to proficiency, or stay fixed?

Factor	Duolingo	Babbel	Busuu	Speak
Speaking Proficiency	Weak ▾ Minimal unprompted production	Weak ▾ Little speaking practice	Weak ▾ Reads a scripted phrase aloud; the target sentence stays visible throughout, so it's recognition, not recall	Strong ▾ Built around conversational production
Tone	Strong ▾ Colloquial tone drives "best-in-class engagement"	Strong ▾ Professional, learning-first	Weak ▾ "Less entertaining, harder to build habit"	Strong ▾ AI-adjusted tone
Contextual Vocabulary	Moderate ▾ Simple vocabulary, but not sequenced to real-world scenario	Strong ▾ Tourist-phrase sequencing early	Weak ▾ Vocabulary is limited and based on frequency, not sequenced to real-world scenarios	Moderate ▾ Teaches conversational vocabulary through speaking prompts, but doesn't sequence it by real-world scenario
UI	Strong ▾ Single clear CTA per screen, minimal navigation complexity	Moderate ▾ Plain interface, low friction	Moderate ▾ Plain interface, low friction	Moderate ▾ Plain interface, low friction
Grammar Explanations	Moderate ▾ "Explain My Answer" gives grammar feedback after an incorrect response, but doesn't explain the underlying rule	Strong ▾ Accurate rule-teaching	Weak ▾ No meaningful grammar explanation	Moderate ▾ Corrects errors in real time, but doesn't teach the underlying rule explicitly

Repetition	Strong ▾ “Let’s recap your errors” at the end of each lesson, and beginning of some	Moderate ▾ Some adaptivity	Weak ▾ Repetition doesn't adjust to demonstrated proficiency	Strong ▾ AI-adapted to proficiency
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Takeaway: Duolingo scores Strong on the three dimensions that drive engagement and habit formation, Tone, UI, and Repetition, confirming it's genuinely best-in-class there. It scores Moderate on Contextual Vocabulary and Grammar Explanations, each offering partial functionality without fully closing the gap. The one clear weakness is Speaking Proficiency, the single dimension that most directly predicts whether a user can produce language unprompted. That's the actual finding: Duolingo needs a production-focused mechanic addressing the one dimension where it's genuinely behind.

B: Personas

Primary: The "Pattern-Recognized" Intermediate. 1,000+ day streak holders who've plateaued. Complete lessons nightly to protect streak and rank, rely on visible cues to answer correctly, and avoid speaking features that feel unstructured or paywalled. Friction: freeze in real conversation despite high in-app accuracy; describe lessons as repetitive "maintenance" rather than learning.

Secondary: The "High-Stakes" Academic Student. Uses Duolingo alongside IB/AP coursework. Friction: app vocabulary doesn't match exam syllabus; no feedback on whether an answer would satisfy a grading rubric, leading to fragmented practice across Duolingo, past papers, and ChatGPT.

C: RICE Prioritization

Problems

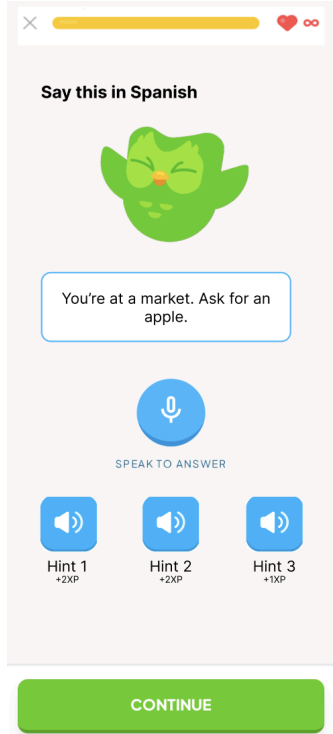
Problem	Reach	Impact	Confidence	Effort	Score
Recognition rewarded over production	5	5	5	3	41.7
Streak-chasing over learning	5	4	4	2	40
Speaking features gated behind subscription	3	4	5	2	30
Vocabulary not scenario-relevant	4	4	4	3	21.3
Lessons feel repetitive	4	3	4	3	16
No feedback on spoken errors	3	4	4	4	12
XP doesn't track exam readiness	2	5	3	5	6

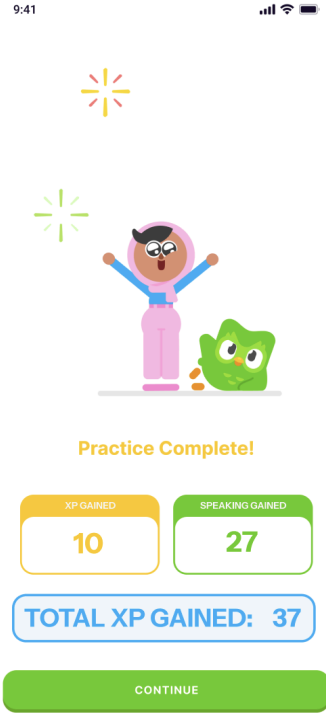
Solutions

Problem	Reach	Impact	Confidence	Effort	Score
Speaking Retention Practice (no visible target sentence)	5	4	5	1	100
Speaking XP Visibility (additive, not replacing existing scores)	5	4	5	1	100
Real-World Survival Modules	4	4	4	3	21.3
Free Daily AI Mini-Conversation	4	4	4	3	21.3
AI Voice Correction	3	5	4	4	15
Smart Repetition (skip mastered content)	4	3	4	4	12
Academic Rubric Grading Mode	2	5	3	5	6

D: Solution Wireframes

The following wireframes illustrate:

Solution 1: Speaking Retention Practice User Flow: • User receives a prompt in English (no visual aids): <i>"Ask for an apple."</i> • Expected response: <i>"Yo quiero una manzana."</i> • User responds verbally in Spanish. • Correct response → +3 XP • Hint 1 (noun cue: "Man-") → +2 XP • Hint 2 (verb cue: "Qui-") → +2 XP • Hint 3 (full sentence shown) → +1 XP Encourages active recall by requiring users to produce language without visual cues, bridging the gap between recognition and real conversation. Tiered hints maintain challenge while offering support, and weighted XP rewards prioritize effortful speaking, improving long-term retention and confidence.	
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Solution 2: Speaking XP Visibility User Flow: - User completes a lesson - Completion screen breaks down XP earned by activity type (speaking vs. reading/listening/typing) - Speaking XP is highlighted to emphasize its higher value Makes speaking progress explicit, reinforcing its importance and motivating users to prioritize productive skills that translate to real-world communication.	
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E: Success Metrics & 30/60/90

Metric Type	Name	Definition	Why?
North Star	Speaking Success Rate	% of speaking tasks completed correctly without hints	Direct measure of unprompted production
Secondary	First-Try Recall Rate	% correct on first attempt, no hints or retries	Signals actual recall vs. eventual guessing
Secondary	Time-to-First-Response	Avg time to begin speaking after a prompt	Proxy for recall fluency
Guardrail	Hint Usage Rate	% of exercises relying on hints	Flags regression toward recognition-based answering
Guardrail	Speaking Task Abandonment Rate	% of users exiting before completing a speaking task	Flags excessive friction

30 days: watch Speaking Task Abandonment Rate. If users are dropping out of speaking exercises at a high rate, friction has exceeded the habit, and the difficulty curve needs revisiting before scaling further.

60 days: watch First-Try Recall Rate. Improvement here is the earliest sign the mechanic is building real recall rather than just tolerance for discomfort.

90 days: Speaking Success Rate should show a statistically meaningful lift over baseline. If it hasn't moved, the hypothesis that XP-weighting alone changes behavior is likely wrong, and the intervention needs to be more structural.

F: Omitted Ideas

- **Dual leaderboards (speaking vs. XP):** merged into a single leaderboard with speaking weighted higher, since separate boards would let users default to whichever is easier to climb.
- **Forced peer-to-peer speaking:** dropped – live, unmoderated speech raises both performance-anxiety-driven abandonment and harassment risk.
- **Hard grammar modules:** dropped for active production practice. Competitor research showed textbook-style grammar teaching feels "aggravating" as opposed to Duolingo's colloquial tone.
- **Standalone academic rubric grading mode:** deferred rather than built as a v1 feature. Duolingo's existing "Explain My Answer" already gives grammar feedback on incorrect responses; a rubric-specific layer that maps answers to exam-specific criteria (e.g., IB/AP structural requirements) is a narrower, later-phase addition once the core production mechanic is validated, and not a replacement for what already exists.